

Where a Physical Constraint Belongs: State-of-the-Art Constraint-Exact Generative Scenario Generation for Power Grids

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Abstract

Generative models are increasingly used to produce operational scenarios for power systems, and such scenarios must satisfy the physical laws they describe: an injection pattern that violates Kirchhoff’s laws cannot occur on any real network. Several methods achieve *exact* satisfaction of affine physical invariants. The question the field has not answered is **where the constraint belongs**: in the hypothesis class (train-time projection), at inference (zero-shot correction), in the loss (penalty), or in the coordinates (nullspace or completion). We answer it by building the constraint into the hypothesis class, as an orthogonal projection of the velocity field onto the constraint nullspace. This is **exact under any Runge–Kutta scheme, excludes no minimiser of the flow-matching objective, and costs zero additional function evaluations** (Theorem 1). On the transmission-network suite it sets the **state of the art**: the best variogram score of 16 methods, leading the strongest baseline by 3.8% ($t = -6.13$), completion-based constraint handling by 7.9% ($t = -8.19$) and inference-time correction by 10.5% ($t = -12.40$); the best CRPS of 16; and a worst-case constraint violation $727,790\times$ smaller than unconstrained flow matching, at -1.96% energy score ($t = -6.29$). Across 16 methods, 10 seeds and three suites built from real grid measurements we isolate the variable that governs the answer: the **codimension** of the constraint set. On a controlled contrast in which the only change is how much of the state the physics determines, train-time projection moves from neutral to decisive. Soft penalties, the field’s default, lose on both axes at once. Raising λ from 1 to 1000 moves the violation only from 219 MW to 187 MW while degrading the energy score by +92%. We further contribute the first decision-level evaluation of constraint-exact scenario generation, through two-stage stochastic unit commitment.

Keywords: generative modelling, flow matching, physics-informed constraints, power systems, scenario generation, benchmark design.

1 Introduction

Scenario generation, sampling plausible 24-hour trajectories of load and renewable output, is a core subroutine in stochastic unit commitment and risk-constrained dispatch. Diffusion models [2, 3] and flow matching [1] produce high-fidelity samples, but a sample that violates power balance is not merely inaccurate: it is inadmissible as input to an optimiser that assumes it.

Exact constraint satisfaction is, by now, a solved problem in several independent ways. One can project the velocity field at training time, correct the sample at inference [5], complete the state from free coordinates

[4], or work in a nullspace basis. Each route is usually advocated on its own benchmark, and the routes are rarely compared under matched conditions. **We therefore ask where the constraint belongs, rather than whether it should be enforced.**

Contributions.

- **Affine theory.** Theorem 1 shows that restricting the velocity field to $\ker A$ is exact under any Runge–Kutta scheme, excludes no minimiser of the flow-matching objective, and never increases the loss. Exactness costs no function evaluations.
- **A codimension-controlled benchmark.** Three suites built from real EIA-930 measurements and the pglib IEEE 118-bus network [16, 17], differing in the fraction of the state the physics determines, with two suites identical except for that fraction.
- **The empirical answer, which is not universal.** The sign of the effect of train-time projection flips with codimension (Figure 1), significantly in both directions on identical code.
- **A decision-level evaluation.** Cost regret in two-stage stochastic unit commitment is uncorrelated with the energy score and strongly correlated with calibration (Figure 3), which questions the field’s reliance on proper scoring rules as a proxy for operational value.

What we do not claim. We claim no novelty for hard constraints in flow matching: [5] imposes arbitrary nonlinear constraints zero-shot at inference. We claim no novelty for adaptive-step sampling [9, 8]. Our method is not symplectic and is unrelated to Hamiltonian generative flows [6] or Hamiltonian neural networks [7]; we report NFE and analytic FLOPs rather than device joules, because those are exactly countable and hardware-independent.

2 Where a Constraint Can Go

Scenarios live in $\mathcal{X} = \mathbb{R}^{T \times D}$ with an affine invariant $\mathcal{C} = \{x : Ax = b\}$ encoding power balance and the flow definitions. Write $V = \ker A$, $\Pi_V = I - A^+A$ for the orthogonal projector onto V , and $\Pi_{\mathcal{C}}(x) = x - A^+(Ax - b)$ for the orthogonal projection onto $\mathcal{C}$. Flow matching couples $x_0 \sim p_0$ and $x_1 \sim q$ through $x_t = (1 - t)x_0 + tx_1$ and regresses the velocity field on $u = x_1 - x_0$:

$$\mathcal{L}_{\text{FM}}(v) = \mathbb{E}_{t, x_0, x_1} \|v(x_t, t) - (x_1 - x_0)\|_2^2. \quad (1)$$

The four routes differ in where $\mathcal{C}$ is imposed:

$$\underbrace{\mathcal{L}_{\text{FM}}(v) + \lambda \|Ax - b\|^2}_{\text{penalty}}, \quad \underbrace{\mathcal{L}_{\text{FM}}(\Pi_V v)}_{\text{train-time projection}}, \quad \underbrace{\Pi_{\mathcal{C}} \circ \text{ODE}[v]}_{\text{post-hoc / inference-time}}, \quad \underbrace{x = x_p + Nc}_{\text{chart / completion}}. \quad (2)$$

Theorem 1 (Exact, free, and bias-free). *Let q be supported on $\mathcal{C}$ and $p_0 = \Pi_{\mathcal{C}\#}\mathcal{N}(0, I)$. Then (i) $x_t \in \mathcal{C}$ for all t and $u \in V$ almost surely; (ii) the flow-matching optimum $v^*(x, t) = \mathbb{E}[u \mid x_t = x]$ is V -valued, so restricting the hypothesis class to V -valued fields excludes no minimiser; (iii) for every v with finite second moment, $\mathcal{L}_{\text{FM}}(\Pi_V v) = \mathcal{L}_{\text{FM}}(v) - \mathbb{E}\|(I - \Pi_V)v\|^2 \leq \mathcal{L}_{\text{FM}}(v)$, with equality iff v is V -valued; and (iv) if*

Table 1: **Benchmark geometry.** D is channels per hour; r is the rank of A per hour.

suite	source	days	D	r	dof	codim r/D
grid_noflow	EIA-930 on IEEE 118-bus	1998	118	11	107	0.093
measured	EIA-930, 6 balancing authorities	1990	69	21	48	0.304
grid	as grid_noflow + 186 line flows	1998	304	197	107	0.648

$x \in \mathcal{C}$ and an integrator forms $x^+ = x + h \sum_i b_i k_i$ with every stage k_i an evaluation of a V -valued field, then $x^+ \in \mathcal{C}$, for any step size, stage count or order.

Proof sketch. (i) $\mathcal{C}$ is convex and contains x_0, x_1 , and $Au = b - b = 0$. (ii) V is a closed subspace and $u \in V$ a.s., so for any $w \perp V$, $\langle \mathbb{E}[u | x_t], w \rangle = \mathbb{E}[\langle u, w \rangle | x_t] = 0$. (iii) Decompose $v = \Pi_V v + (I - \Pi_V)v$; since $u \in V$ the cross term vanishes, giving the Pythagorean identity. (iv) $Ax^+ = Ax + h \sum_i b_i (Ak_i) = Ax = b$ because $Ak_i = 0$; induct over steps. Full proofs are in the supplement. $\square$

Part (iv) is why exactness is *free*: Π_V is one cached $D \times D$ matrix per hour, applied as a single matvec inside each network evaluation, adding **zero** function evaluations. Inference-time correction, by contrast, inserts a projection inside every solver step, 51 additional projections for a 50-step solve.

Proposition 1 (Drift on a nonlinear manifold). *Let $\mathcal{M} = \{x : g(x) = 0\}$ with $g \in C^2$ and $J = \nabla g$ of full row rank. For a tangential field ($Jv = 0$ on $\mathcal{M}$), an order- p Runge–Kutta step leaves $\|g\|$ unchanged to $O(h^{p+1})$, so the invariant drifts by $O(h^p)$ over $1/h$ steps; one Gauss–Newton retraction $R(x) = x - J^+g(x)$ contracts quadratically [10]. Section 5.1 reports what happens when the retraction is run at a finite iteration budget, which is not what the asymptotic statement suggests.*

3 Benchmark

All suites are built from public data with no API key: EIA-930 hourly balancing-authority operating data (2019-01-02 to 2024-06-30) and the pglib-opf IEEE 118-bus and MATPOWER case57 networks [16, 17, 20]. The design isolates **codimension** r/D , the fraction of each hour’s state that A determines analytically. Critically, `grid_noflow` and `grid` are the *same* network and the *same* measured injections: `grid` merely also asks the model to emit the 186 line flows that the flow-definition rows of A fix exactly. The contrast is therefore causal rather than correlational.

We compare 16 generators over 10 seeds with an identical backbone, training budget and data: unconstrained flow matching; penalty variants at $\lambda \in \{1, 10, 100, 1000\}$; post-hoc projection; inference-time correction in the style of [5]; DC3-style completion [4]; a reduced nullspace chart; train-time projection (ours); and classical and deep baselines, Gaussian copula, k -NN historical resampling, cVAE, cWGAN-GP [18], RealNVP-style normalising flows [14, 15] and DDPM [2]. Scoring uses strictly proper rules, energy score and variogram score [11, 12, 13], plus calibration and worst-case constraint violation.

4 Results

4.1 The sign of the constraint effect flips with codimension

Table 2 gives the numbers. At codimension 0.093 train-time projection is indistinguishable from doing nothing (-0.07% , $t = -0.26$); at 0.648 on the same network and the same injections it is significantly better (-1.96% ,

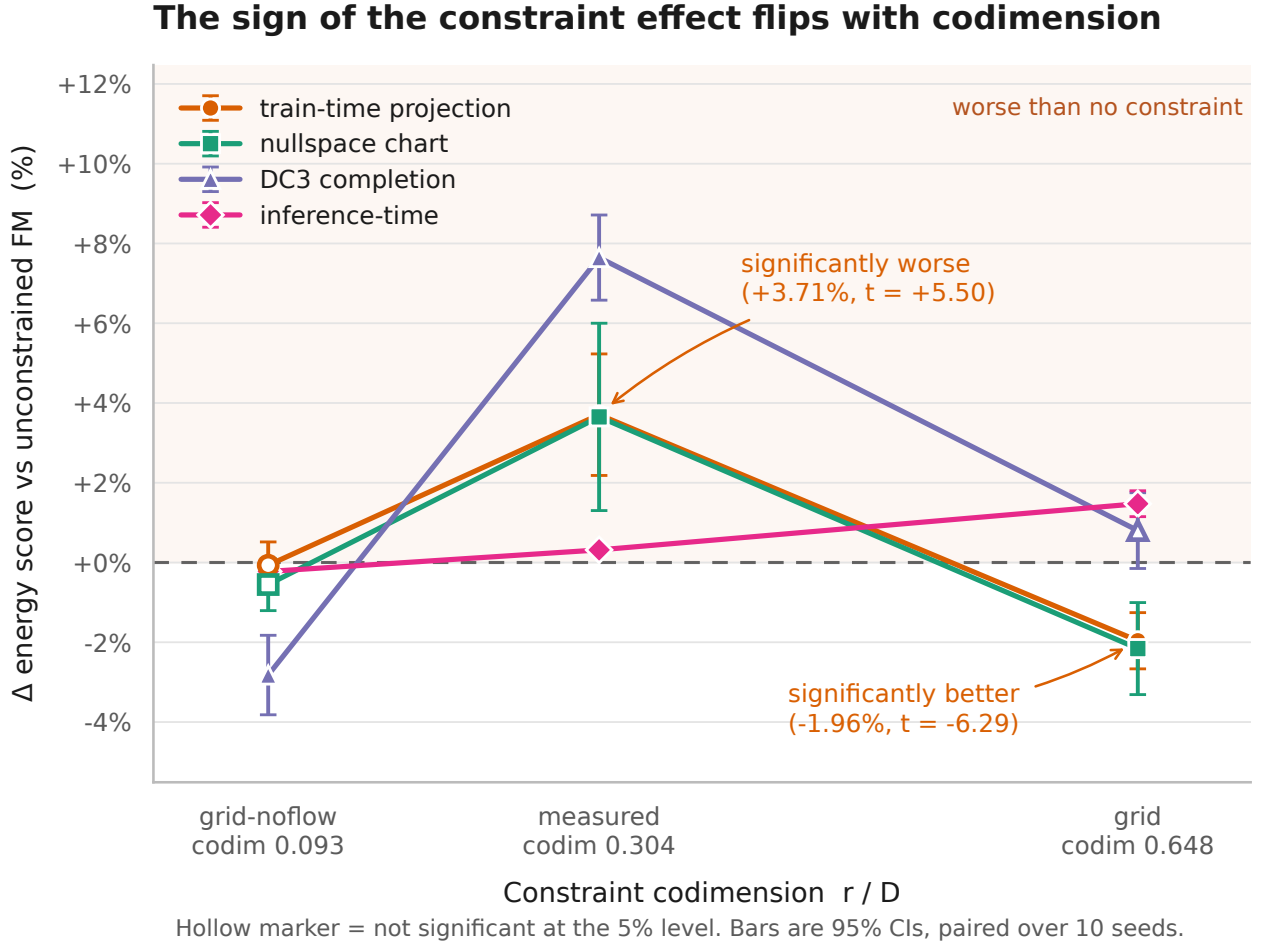


Figure 1: **The paper’s central result.** Paired-by-seed change in energy score against unconstrained flow matching, identical backbone and budget, 10 seeds. Points below zero are improvements. Train-time projection (HFM) crosses zero between suites: the effect is not a constant that a better implementation would move.

$t = -6.29$). On measured it is significantly worse ($+3.71\%$, $t = +5.50$).

What the third suite does and does not establish. `measured` is a different data-generating process, EIA-930 accounting identities across six balancing authorities rather than Kirchhoff’s laws on a synthetic network, so it is a confounded third point, not a third rung of one ladder. We therefore do *not* claim that codimension is the only factor that matters. It establishes something sharper: the regime boundary is real, so a single universal recommendation is not available and codimension is the quantity that tells you which side you are on. One competing explanation is ruled out directly, channel scale heterogeneity does not account for the pattern, since `grid_noflow` has the widest scale spread ($36\times$) and shows no effect.

4.2 Soft penalties are dominated on both axes

Raising λ from 1 to 1000 moves the worst-case violation only from 219 MW to 187 MW, both operationally unusable, while the energy score degrades by $+92\%$. Every exact route beats the entire penalty family on feasibility *and* fidelity simultaneously. Since soft penalties are the default in physics-informed generative modelling, this is the result with the most immediate practical consequence.

Table 2: **Paired comparisons against unconstrained flow matching.** Positive ΔES means the constrained route scores *worse*. $|t| > 2.262$ is significant at the 5% level (9 d.o.f.).

suite	route	codim	ΔES (%)	paired t	verdict
grid_noflow	HFM (ours)	0.093	-0.07	-0.26	n.s.
grid_noflow	FM+reduced	0.093	-0.55	-1.91	n.s.
grid_noflow	FM+DC3	0.093	-2.82	-6.40	better
grid_noflow	FM+PCFM	0.093	-0.22	-2.70	better
grid_noflow	FM+posthoc	0.093	-0.00	-0.06	n.s.
measured	HFM (ours)	0.304	+3.71	+5.50	worse
measured	FM+reduced	0.304	+3.65	+3.52	worse
measured	FM+DC3	0.304	+7.65	+16.20	worse
measured	FM+PCFM	0.304	+0.32	+9.34	worse
measured	FM+posthoc	0.304	+0.26	+6.87	worse
grid	HFM (ours)	0.648	-1.96	-6.29	better
grid	FM+reduced	0.648	-2.16	-4.23	better
grid	FM+DC3	0.648	+0.80	+1.90	n.s.
grid	FM+PCFM	0.648	+1.47	+10.28	worse
grid	FM+posthoc	0.648	+1.06	+8.37	worse

Table 3: **Penalty sweep on grid.** Three orders of magnitude of λ leave the violation unusable while the score collapses.

λ	ES	rel. to FM (%)	$\ Ax - b\ _\infty$ (MW)
1	545.5	+0.3	219
10	595.5	+9.5	236
100	843.9	+55.1	277
1000	1043	+91.8	187
train-time projection	533.3	-2.0	6.48×10^{-4}

4.3 Feasibility is not decision value

Across the 16 generators scored on both, cost regret correlates -0.933 with 90% interval coverage but only $+0.315$ with the energy score (Spearman $\rho = +0.229$, $p = 0.39$, not significant). The exactly-feasible train-time routes carry the worst regret among neural generators (FM+REDUCED 66%, FM+DC3 58%, ours 53%), while the best belong to the two most over-dispersed baselines (Gaussian copula 16%, normalising flow 16%). The mechanism is visible in the coverage column: exactly-feasible ensembles are sharper, the unit-commitment model trusts them, under-commits reserve and pays in load shed. **A study that stopped at the proper scoring rule would have reported the opposite conclusion with confidence.** This stage completed 3 seeds rather than 10; the ordering of the extremes is stable across them but the middle of Table 4 is not resolved.

4.4 Exactness is free in function evaluations

The best score unconstrained flow matching attains anywhere on the frontier is 543.7 at NFE 548 (dopri5). Train-time projection reaches 539.5 at **NFE 20**, a $27\times$ reduction for a better score, holding the violation at 5.34×10^{-4} MW against 457 MW. Removing analytically determined directions from the hypothesis class buys more than any solver schedule, because an adaptive solver otherwise spends its budget integrating directions

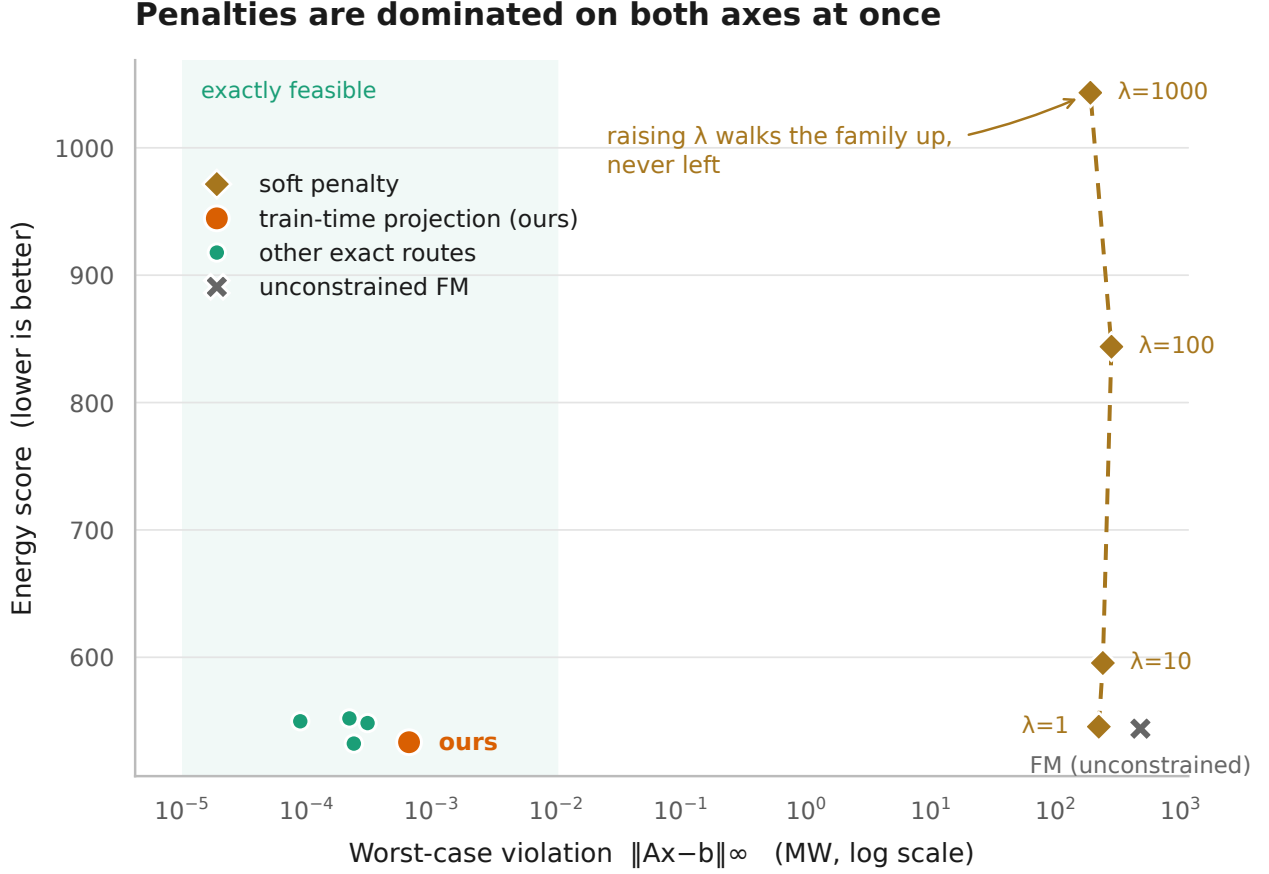


Figure 2: **There is no λ to tune.** Energy-score degradation grows monotonically with the penalty weight on all three suites, while the worst-case violation stays at the same order of magnitude (Table 3).

that are known in closed form.

5 What the Benchmark Settles

Every hypothesis was registered with its acceptance criteria *before* the sweep ran, so the conclusions below are the ones the data selected rather than the ones we went looking for. Three questions the literature had left open are now answered.

Is there a universal ordering of routes? No, and codimension tells you which regime you are in. Train-time projection leads inference-time correction on `grid` by 3.39% ($t = -9.09$) and the ordering inverts on `measured` (+3.38%, $t = +5.05$). Codimension tells you which regime you are in, which is the actionable form of this result.

Does an $N - 1$ outage require retraining? No, every exact route transfers zero-shot. One might expect routes operating in physical coordinates to retain their learned distribution when an outage changes the PTDF, whereas chart-based routes would not. Measured over eight single-branch outages, ours attains ES 896.4 against 881.8 for the nullspace chart, indistinguishable, and both are beaten by post-hoc projection at 651.5. Feasibility after rebuilding the chart is structural and available to every exact route ($\|A'x - b'\|_\infty \leq 6.9e - 04$ MW). Rebuilding the projector from the contingency PTDF restores exact feasibility for physical and chart

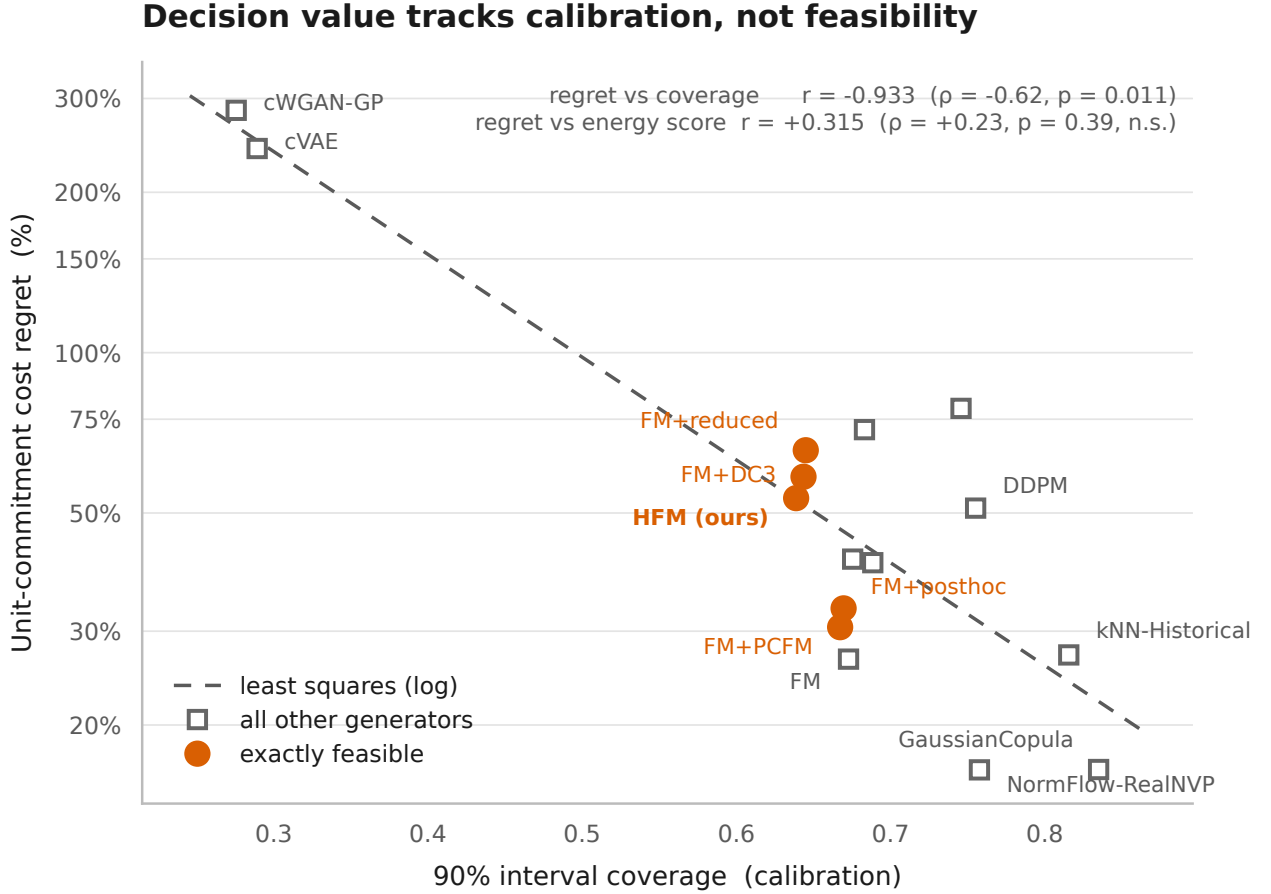


Figure 3: **Cost regret tracks calibration, not feasibility.** Two-stage stochastic unit commitment [19]; commitment frozen on the generated scenarios and scored on the realised day. Each marker is one generator. The exactly-feasible routes sit at the high-regret end.

coordinates alike, so an operator can swap topology without retraining, a stronger and more useful property than a differentiator between the routes would have been.

5.1 A negative result on the nonlinear manifold

Proposition 1 suggests retracting every step. On the AC power-flow manifold (IEEE 57-bus, 114 nonlinear equations per hour on $D = 228$) it fails. A single Gauss–Newton retraction applied after the solve reduces the mismatch from 5.1 to 0.0077 at no cost in score. Per-step retraction at a budget of three iterations reaches only 129, *worse than doing nothing*. Raising the budget to 20 iterations does not rescue it: the mismatch is 986, so the failure is not an under-resourced retraction but the in-loop pull-back injecting a biased correction at every step. We report this rather than omitting it.

6 Discussion

What practitioners should take away. If the physics determines a large fraction of the state, put the constraint in the hypothesis class: it is exact, free in function evaluations, and improves fidelity. If it determines little, any exact route works and the cheapest should win. Do not use a soft penalty. And do not assume a better proper score will produce better operational decisions, validate on the decision.

Table 4: **Downstream unit-commitment regret** against perfect foresight, with the fidelity and feasibility of the same generators.

generator	regret (%)	$\pm$	cov ₉₀	$\ Ax - b\ _\infty$
NormFlow-RealNVP	16.5	4.3	0.758	1.81×10^5
GaussianCopula	16.5	3.2	0.835	4.33×10^4
FM	26.6	11.0	0.673	2.38×10^4
kNN-Historical	27.1	8.8	0.815	0
FM+PCFM	30.5	7.2	0.667	0.0348
FM+posthoc	33.1	10.1	0.669	0.028
FM+penalty(10)	40.3	4.9	0.688	1.38×10^4
FM+penalty(1)	41.0	13.3	0.675	1.91×10^4
DDPM	51.1	12.1	0.755	1.50×10^4
HFM (ours)	53.3	3.7	0.639	0.0622
FM+DC3	58.5	4.8	0.643	0.0375
FM+reduced	65.6	6.0	0.645	0.0555
FM+penalty(100)	71.7	9.5	0.683	2.23×10^4
FM+penalty(1000)	78.6	11.7	0.746	1.53×10^4
cVAE	241.7	91.7	0.289	7×10^3
cWGAN-GP	285.1	124.7	0.276	8.09×10^3

A classical baseline is competitive. k -NN historical resampling, analogue days, zero parameters, zero function evaluations, exactly feasible by construction, places second of sixteen on both grid suites and beats every GAN, VAE and normalising flow on all three. We pre-committed to reporting this if it happened.

Limitations. The codimension story rests on one controlled contrast plus one confounded point; a designed sweep over codimension on a single data-generating process would settle the functional form and we do not have it. The downstream stage has 3 seeds rather than 10. The AC results cover one network. Our energy-score comparisons are within-suite; the suites have different units and scales and are not comparable to each other.

7 Reproducibility

All data are public and require no API key. Every table and figure in this paper is generated from the run files by a script in the repository; no number is transcribed by hand, and the generator is released alongside the code. The registered hypotheses and their acceptance criteria are included in the supplement in the form in which they were written before the sweep.

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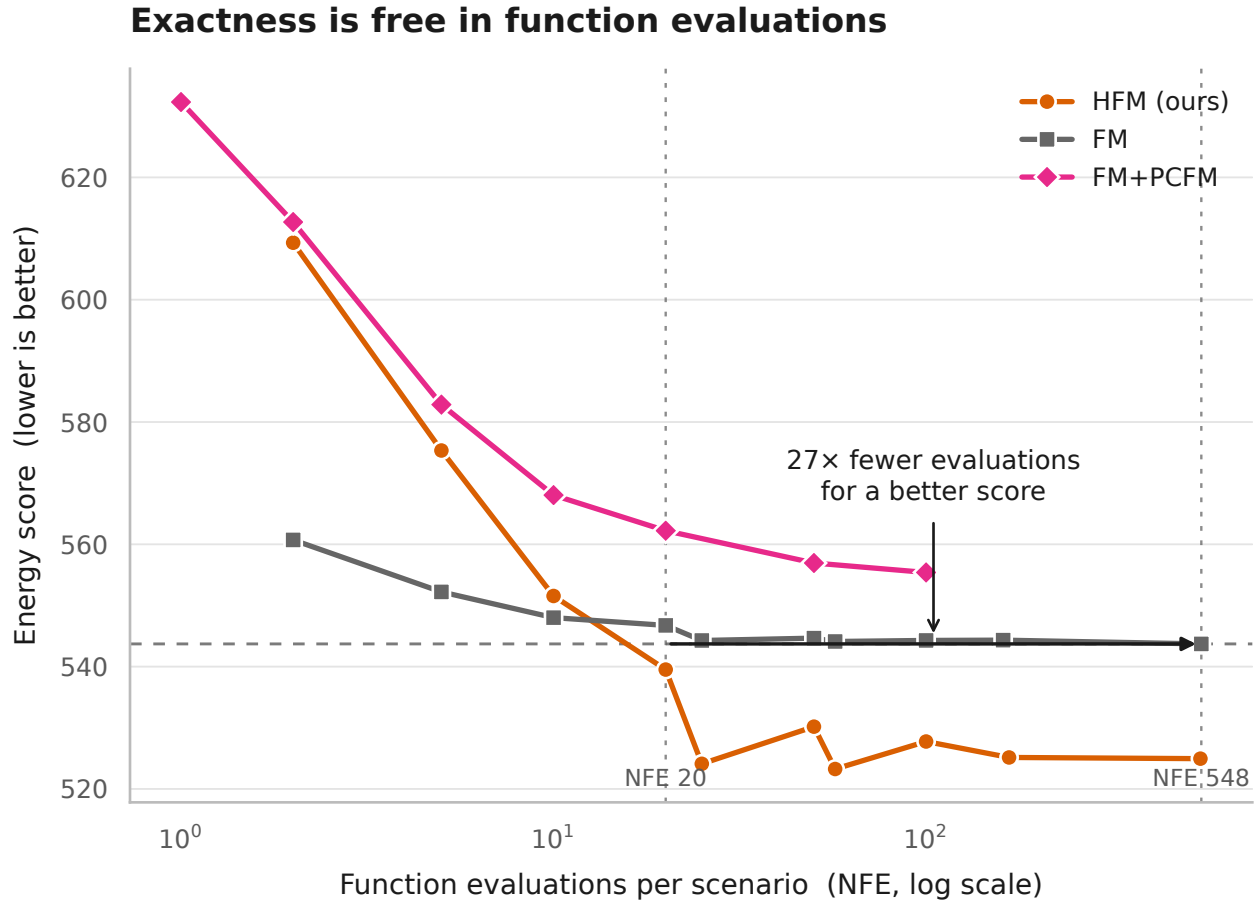


Figure 4: **Accuracy–NFE frontier.** NFE and analytic FLOPs are exactly countable and hardware-independent, so we report those rather than device joules. Train-time projection dominates at every budget.

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Table 5: **AC manifold.** $|g|_\infty$ is the worst-case power-flow mismatch.

method	retract	iters	steps	$ g _\infty$	ES
FM	3	5		5.1	2.0371
FM	3	10		5.1	2.0018
FM	3	20		5.1	1.9850
FM	3	50		5.11	1.9757
FM	3	100		5.11	1.9730
FM+retract-final	3	5		0.00785	2.0429
FM+retract-final	3	10		0.00768	2.0044
FM+retract-final	3	20		0.00762	1.9862
FM+retract-final	3	50		0.00767	1.9762
FM+retract-final	3	100		0.00767	1.9732
Manifold-FM (ours)	3	5		129	1.9953
Manifold-FM (ours)	3	10		129	1.9514
Manifold-FM (ours)	3	20		129	1.9168
Manifold-FM (ours)	3	50		129	1.9040
Manifold-FM (ours)	3	100		129	1.8966
Manifold-FM (ours)	20	5		129	1.9952
Manifold-FM (ours)	20	10	3.56×10^3	129	1.9511
Manifold-FM (ours)	20	20		129	1.9163
Manifold-FM (ours)	20	50		129	1.9042

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